

Real Time Innovations

Tactical Microgrid Example Code Walk-thru

April 9, 2021

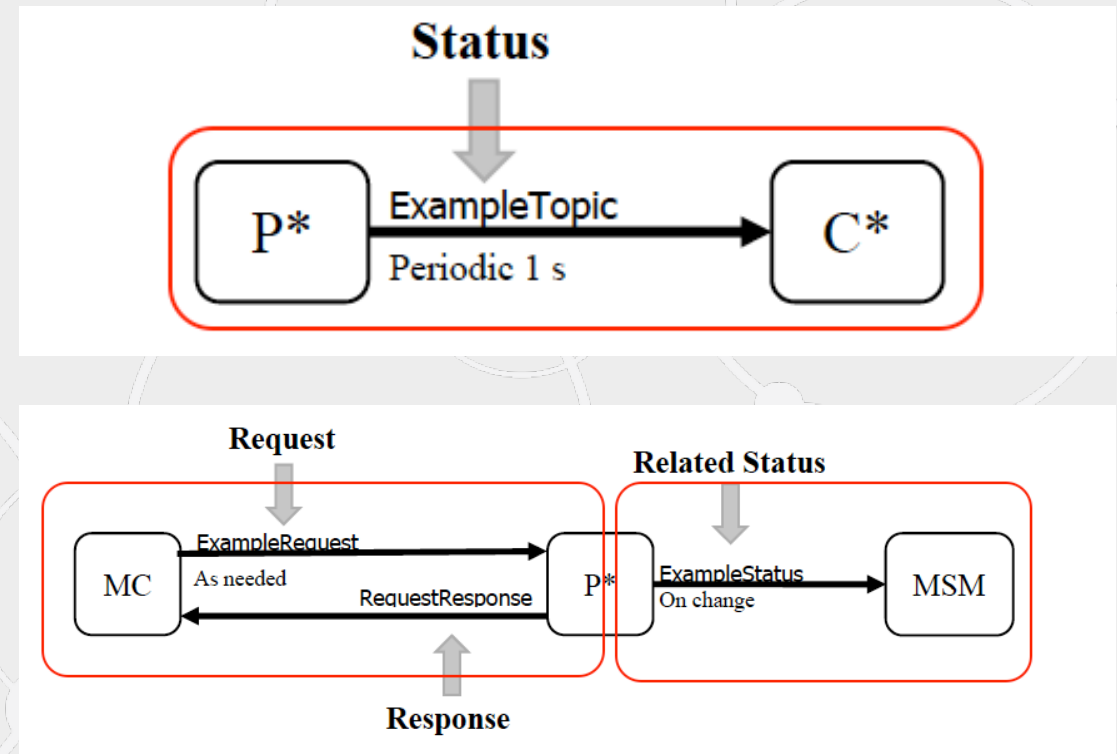
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Revision 1.3

Components Involved

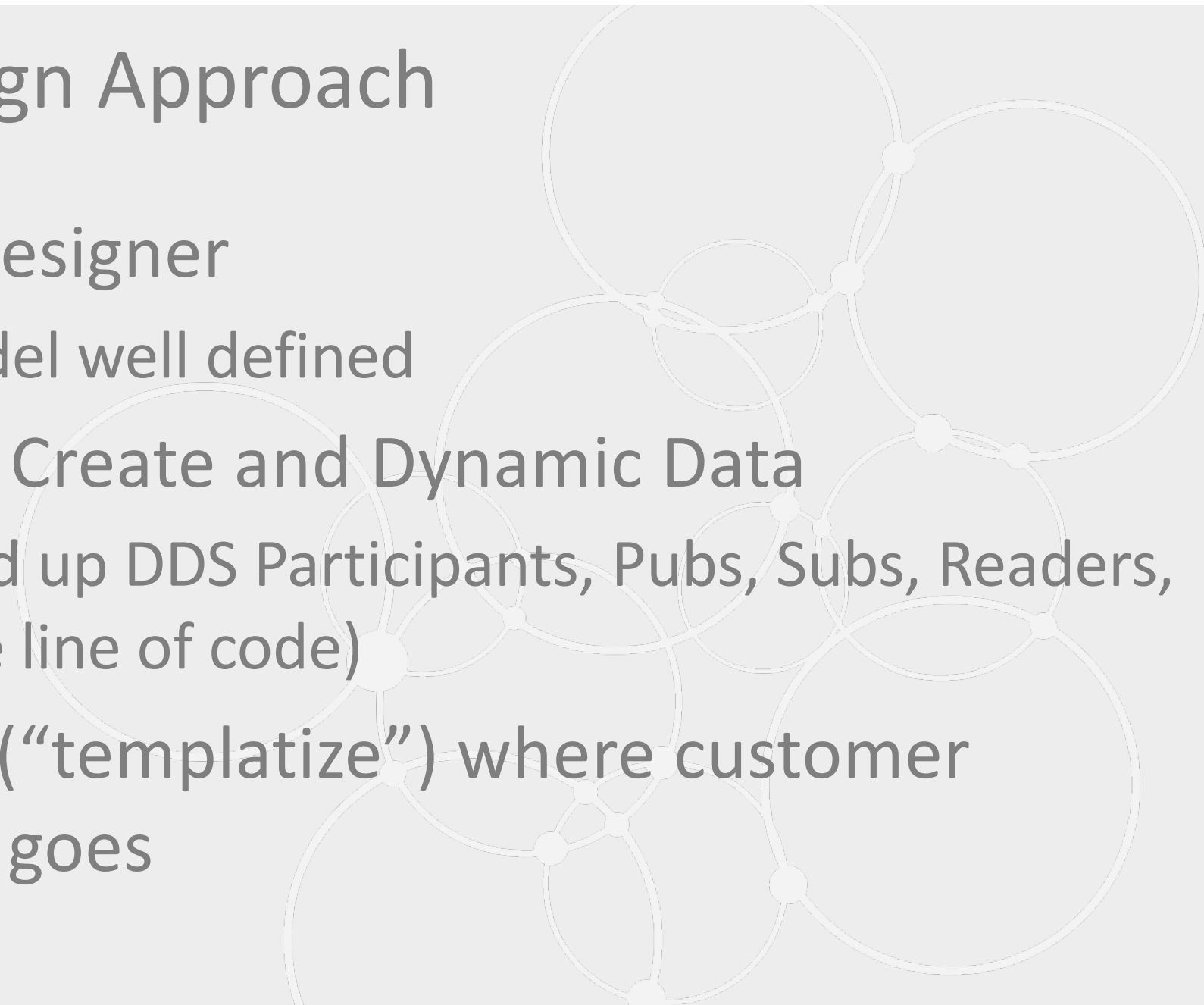
- Tactical Microgrid Spec (TMS)
 - Data Patterns
 - Publish Once (Durable)
 - Periodic
 - Request / Response
 - On-Change
 - Custom (Not implemented)
 - Topics
- Distributed Data Service (DDS)
- Customer Legacy System
 - Gateway Module / Interface between TMS/DDS and Legacy Communications



Issues

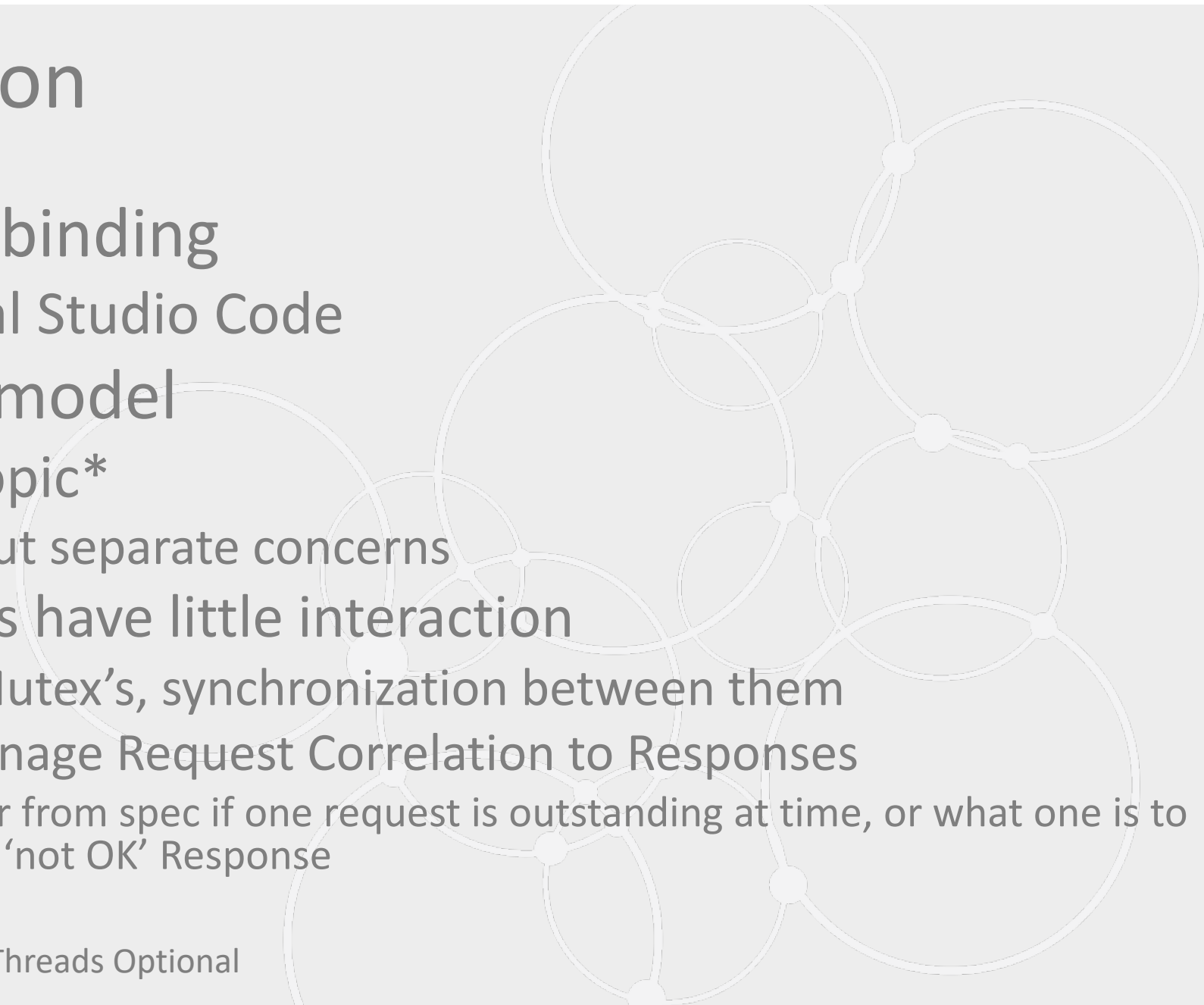
- **TMS Spec is complex** in draft form, and hard to get any examples to help confirm understanding
- **Code Complexity** to provide separation of concerns
- **Lumping Complexities:** TMS and application code structure complexity (to isolate legacy adaptation) may be lumped in as DDS complexities
- **DDS Experience** – recommend RTI Virtual Instructor Led Training (24 hour custom training) to get familiar with DDS

Example Design Approach



- Use System Designer
 - Existing model well defined
- Use XML APP Create and Dynamic Data
 - Easy to stand up DDS Participants, Pubs, Subs, Readers, Writers (one line of code)
- Separate out (“templatize”) where customer adapter code goes

Implementation



- C++ Language binding
 - MacOS, Visual Studio Code
- Thread based model
 - Thread per topic*
 - separates out separate concerns
 - Note: Threads have little interaction
 - No Locks, Mutex's, synchronization between them
 - Rules to manage Request Correlation to Responses
 - Note clear from spec if one request is outstanding at time, or what one is to do with a 'not OK' Response

* Threads per Writer Event Threads Optional

Project Directory Structure

- Directories Under Project Directory tms
 - tmsApp (example Application)
 - tmsAppTest-SimMSM
 - MSM simulation used to unit test tmsApp
- .git – git management directory
- .DS_Store (Mac OS file)
- .gitignore (git file to instruct git to ignore built files)

tmsApp – Directory Structure

- Model (tms Data Model Directory) Contains:
 - Tms supplied idl and xml files
 - tms.idl, tms.xml, tms-topics.idl, tms-topics.xml
 - System Designer Project file 'tmsTestApp.rtisdpj'
 - tmsTestApp.xml file
 - The xml file we'll use to 'standup' all the DDS entities (Participant, Publisher, Subscriber, Readers, Writers and topics)
- cpp – holds source files
- objs – holds the output of the make/build
 - This directory is always removed and recreated with make clean/make (all)
 - It is and its contents are in .gitignor and not checked in the git

tmsApp Directory level files

- makefile – builds tmsApp
- USER_QOS_PROFILES.xml (copy of ./model/tmsTestApp.xml)
 - USER_QOS_PROFILES.xml is the default xml file DDS uses if no URL is specified.
 - We can avoid the copy by setting an env var to direct the application to the non-default URL of “Project Directory path/tmsApp/model/tmsTestApp.xml”
- tms.code-workspace and .vscode directory
 - Visual Studio Code workspace and configuration directories

tmsApp cpp files

- tmsTestExample.h
 - Created by rtiddscode gen from the tmsTestApp.xml file and used by the .cxx files to properly reference tms_ specific data structures and names
- tmsCommon.h
 - my common header file for .cxx files during compilation
- tmsTestExampleApp.cxx
 - This is the application C++ top level code

tmsApp cpp files (con't)

- tmsCommPatterns.cxx & .h files
 - Holds the definitions for the tms communications patterns and associated thread control blocks (see tmsCommPatterns.h file). These include
 - ReaderEventThread
 - Required Thread used for all readers
 - WriterEventThread
 - Optional thread to monitor writer topic status
 - PeriodicWriterThread
 - Use to send periodic topics
 - OnChangeWriterThread
 - Used when the user code sees a change in Status or variable requiring a tms 'On-Change' topic publication
 - Uses a User Defined DDS Guard Condition to trigger the associated topic writer

tmsCommPatterns – Reader Processing

- Control Block (ReaderTheadInfo) option
 - echoResponse (true/false)
 - Intended for received requests that require writing a `tms_TOPIC_REQUEST_RESPONSE`
 - Copies the Request SampleId to the RelatedRequestId
 - Sets the Status code to infoBlock `tms_REPLY_code` and `tms_reason` (can be changed in individual topic handlers)

tmsCommPatterns – Periodic Writer

- Writes associated topic periodically
 - As initialized (passed in variable to) by the control block c'tor
- Control Block (PeriodicTheadInfo) option
 - enabled (true/false)
 - Disables the both the call to the handler and the write operation

tmsCommPatterns – OnChange Operation

- Waits on your user defined DDSGuardCondition
 - Handed in to the associated info block at creation (of the info block) – part of the c'tor
- Idea is change an internal_variable different from the tms_external_variable to cause 'On Change' publishing
 - Due to an internal device change of state or from a tms_request topic
 - In the main_loop, if the internal_variable != tms_external_variable, set up the topic change the condition variable to true
 - Also take care to either preconfigure the topic data (if static) or first change any of the topic data prior to triggering the condition

tmsCommPattern – Topic Handlers

(tmsCommPatternTopicHndlr.cxx & .h)

- Allows customization for topics by user
- Currently only envisioned for all Readers and PeriodicWriters
 - Periodic data is set per specific topic
- No perceived use cases for WriterEvents and OnChange Writers
 - WriterEvents publishes nothing and allows you to ‘handle’ status of the writer changing
 - OnChange you set up the data in the context of where you trigger the event through a DDSGuardCondition
 - Easy to add infrastructure for these if use case arises
 - i.e., for some reason you want to handle a the same writer status differently for one topic vs. another.

tmsCommPattern – Topic Handlers (Con't)

(tmsCommPatternTopicHndlr.cxx & .h)

- Declare a custom handler in tmsCommPatternTopicHndlr.h
- Write a custom handler in tmsCommPatternTopicHndlr.cxx
- Install the handler in the appropriate handler array location in tmsCommPatternTopicHndlr.cxx
 - reader_handler_ptrs[]
 - Periodic_handler_ptrs[]
- The tmsCommsPattern thread processing does the generic/common processing before and after calling the handler
 - E.g., if on a request receive topic, if EchoResponse enabled, the readerThread will
 - First: Detect data arrival, get the data
 - Second: call the handler (passing a pointer to the info block)
 - Third: If EchoResponse enabled, copy the deviceId and sequenceNumber to the relatedRequestId in requestResponse. Copy the handler set or default status code and status reason to the requestResponse and publish

tmsAppTest-SimMSM – Directory Structure

- Similar to tmsApp Directory Structure but...
 - No model directory
 - Uses the Same model dir as tmsApp
 - Has its own Participant defined in the model
- cpp directory only contains the tmsAppTest-SimMSM.cxx
 - Make file builds and links with the same header files, tmsCommPatters and handler files as in the tmsApp directory
 - tmsCommsPatternTopicHndrs.cxx/.h will hold both tmsApp and tmsAppTest-SimMSM handlers
 - But each instantiates complementary topics and patterns

Current Status

- <https://github.com/psmass/tms>
- Provides tmsApp and tmsAppTest-SimMSM
 - MSM Sim test code only implemented enough to keep app stimulated
 - E.g., does not even look at device discovery topics
- Six of eight planned topics basically working
 - Between the App and Test App – all planned patterns working

What's left for Example (priority)

- Dispose of topics once processed
 - Currently TMS implements SampleID as Key'd but each one is unique (device ID + Unique Sequence ID) – this effectively consumes resources for each Topic
- Test Request from Device / Response from MSM correlation mechanism
- Add SourceTransitionRequest to provide example of request to device and OnChange response
 - (currently you can see this from MSM code)

What's left clean up

- Add a callback from Req/Response correlation mechanism if outstanding request is to be overwritten/missed.